

Paper 30 - Grand Ribbon Meta-Framer

CQE-paper-30

CQECMPLX Formal Paper Series

Review-facing PDF built 2026-06-13

Construction Basis

Show the thirty-paper corpus as a ribboned local-rule presentation that prepares Paper 31 without making Paper 31 part of the proof stack.

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-30.md
- paper kernel manifest: production\paper-kernels\papers\CQE-paper-30\paper_kernel_manifest.json
- verifier: production\formal-papers\CQE-paper-30\verify_grand_ribbon_meta_framer.py
- receipt: production\formal-papers\CQE-paper-30\grand_ribbon_meta_framer_receipt.json (CQE-paper-30: pass_with_open_packaging_obligations)
- body: production\papers\CQE-paper-30\PAPER-BODY.md
- source: production\papers\CQE-paper-30\SOURCE.md
- formal: production\papers\CQE-paper-30\01-CQE-formal\FORMAL.md
- tool: production\papers\CQE-paper-30\02-CQE-tool\TOOL.md
- workbook: production\papers\CQE-paper-30\03-CQE-workbook\WORKBOOK.md
- recovered_output: production\lib-forge\recovered\papers_output\CQE-paper-30.md

Paper 30 - Grand Ribbon Meta-Framer

Abstract

Paper 30 frames papers 00-29 as one swept local-rule ribbon. Each paper is not treated as an isolated artifact for this purpose; it is a position in the same eight-slot structure: `C, L, R, B, T, O, W, A`, meaning center, left boundary, right boundary, boundary rule, tool transform, obligation set, workbook analogue, and citation/provenance anchor.

The closed result is structural and bounded. The verifier proves that the eight-slot fill discipline is coherent, that papers 00-29 can be swept as 30 filled ribbon positions with provenance, that the live terminal composition tree returns a single canonical route, and that the transport-obligation ledger passes while keeping open lifts visible. Paper 31 is prepared as the readout of the sweep, but it is not used as a proof dependency for papers 00-29.

Closed Evidence

The receipt is generated by ``production/formal-papers/CQE-paper-30/verify_grand_ribbon_meta_framer.py``. It writes ``grand_ribbon_meta_framer_receipt.json``.

The first closed row is the ribbon schema. The slot order is:

C, L, R, B, T, O, W, A

A slot is filled only when it has both a value and a provenance entry. The allowed source kinds are ``binary``, ``vector``, and ``binary+vector``.

The second closed row is the corpus sweep. The verifier builds 30 ribbon positions, from ``CQE-paper-00`` through ``CQE-paper-29``, and fills all eight slots at every position. The left and right slots bind each paper to its prior and next position in the proof sweep; the center slot binds the paper's claim center; the remaining slots bind the readout law, tool transform, obligations, workbook supplement, and provenance anchors.

The third closed row is the live terminal route. The verifier builds the seed database and calls ``build_terminal_composition_tree`` for ``Niemeier:A2^12``. The summary returns a generated canonical composition tree with a single canonical route after component ordering and orbit quotient.

The fourth closed row is the transport boundary. The verifier calls ``verify_transport_obligations``, which returns ``pass_with_open_lifts``. This is important: the paper proves that the boundary ledger is visible and usable; it does not pretend that every lift is already demonstrated.

Definitions

The active center ``C`` of a paper is the claim center selected by that paper's observer event. In this paper, all 30 centers are read as positions on one swept ribbon.

The left and right boundaries ``L`` and ``R`` are the predecessor and successor positions in the production proof sweep.

The boundary rule ``B`` is the local LCR/Rule 30 readout discipline inherited from the earlier papers.

The tool transform ``T`` is the verifier, package surface, or formal receipt that the paper uses to make its claim replayable.

The obligation slot ``O`` is the open-residue set. It is filled when obligations are named, not when they disappear.

The workbook slot ``W`` is the analog or quarter-step supplement that exposes the same state without requiring a software stack.

The anchor slot ``A`` records citation and provenance anchors.

Claims

- A valid paper ribbon has exactly eight slots in the order ``C, L, R, B, T, O, W, A``.
- A slot is accepted only when both value and provenance are present.
- Papers 00-29 form a 30-position proof sweep under this slot discipline.
- The live terminal tree supplies a single canonical composition route for the tested terminal.
- The transport ledger is part of the ribbon boundary and must preserve open lifts as open.
- Paper 31 is a retrospective readout of the sweep, not a premise needed by papers 00-29.

Theorem 30

The production corpus through Paper 29 can be represented as one provenance filled eight-slot ribbon sweep. This representation is valid exactly when each position fills the eight slots with provenance, the sweep uses papers 00-29 only, the terminal route is canonical, and the transport ledger preserves open lifts instead of hiding them.

Proof

Run ``verify_grand_ribbon_meta_framer.py``.

The slot-schema check passes because the verifier defines the ordered slot set ``C, L, R, B, T, O, W, A`` and rejects source kinds outside the bounded set ``binary``, ``vector``, and ``binary+vector``.

The sweep check passes because the verifier constructs one ribbon for each paper id from ``CQE-paper-00`` through ``CQE-paper-29``. Each position fills all eight slots, and each slot carries a source path as provenance. This proves the paper's structural claim: the corpus can be read as one repeated local form.

The terminal-route check passes because the live ``lattice_forge`` terminal tree returns a generated canonical composition tree and reports a single canonical route after component ordering and orbit quotient. This gives the paper a concrete spine rather than a purely narrative ordering.

The transport-ledger check passes because ``verify_transport_obligations`` returns ``pass_with_open_lifts``. The open lifts are not treated as failures of the ribbon; they are the named boundary residue that keeps later claims honest.

Finally, the dependency check passes because `CQE-paper-31` is not included in the 00-29 sweep. Paper 31 may read the sweep after the fact, but the earlier proof stack does not depend on that readout. Therefore Theorem 30 holds.

Worked Example

Take `CQE-paper-29`. Its center slot is the Monster horizon claim. Its left slot points to Paper 28, and its right slot points to Paper 30. Its boundary rule is the same LCR readout discipline. Its tool transform is the Paper 29 verifier. Its obligation slot contains the quarantined energy-bound hypotheses. Its workbook slot points to the 29.25/29.50/29.75 supplements, and its anchor slot points to the source and receipt provenance.

Now repeat that same slot fill for every paper from 00 through 29. The result is not thirty unrelated objects; it is one form swept across thirty centers. The sweep is valid because every slot is filled with provenance and because the transport boundary remains visible.

Open Obligations

The reusable `cqengine.ribbon` module exists in neighboring kernel and production copies, but it is not yet packaged in this git-hosted production root. This paper's verifier mirrors the contract and records promotion of that module as a packaging obligation.

Legacy workbook language sometimes says "31 beads." Production Paper 30 uses the 30-position proof sweep `00-29` plus Paper 31 as readout. Any future 31-bead display must mark Paper 31 as readout, not as a backward dependency.

The transport ledger still has open lifts. This paper requires those open lifts to remain visible until their own verifiers close them.

Suite Role

Paper 30 is the hinge between proof papers and meta-reading. It gathers the proof stack into one inspectable ribbon without converting the meta-reading into evidence for earlier claims. Paper 31 may now read the corpus as an enacted LCR process, and Paper 32 may package the suite with this ribbon as a selector and status map.